



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Tools Used



Project Overview

Business Background

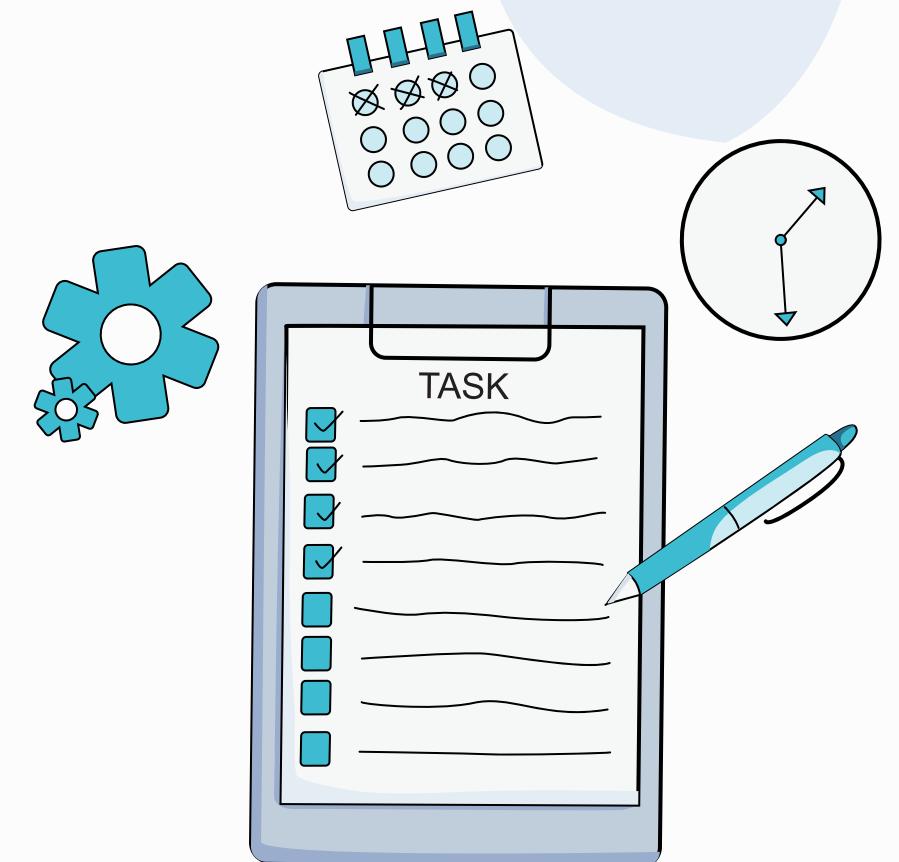
Retail businesses generate thousands of customer transactions every day, but raw transaction data alone provides limited business value. Understanding customer purchasing behavior, spending patterns, and shopping preferences is essential for improving marketing strategies, increasing customer retention, and maximizing revenue. By analyzing customer demographics, purchasing history, product preferences, and subscription behavior, businesses can make more informed, data-driven decisions.

Business Objective

Analyze customer shopping behavior to identify purchasing patterns and generate actionable business insights.

Project Scope

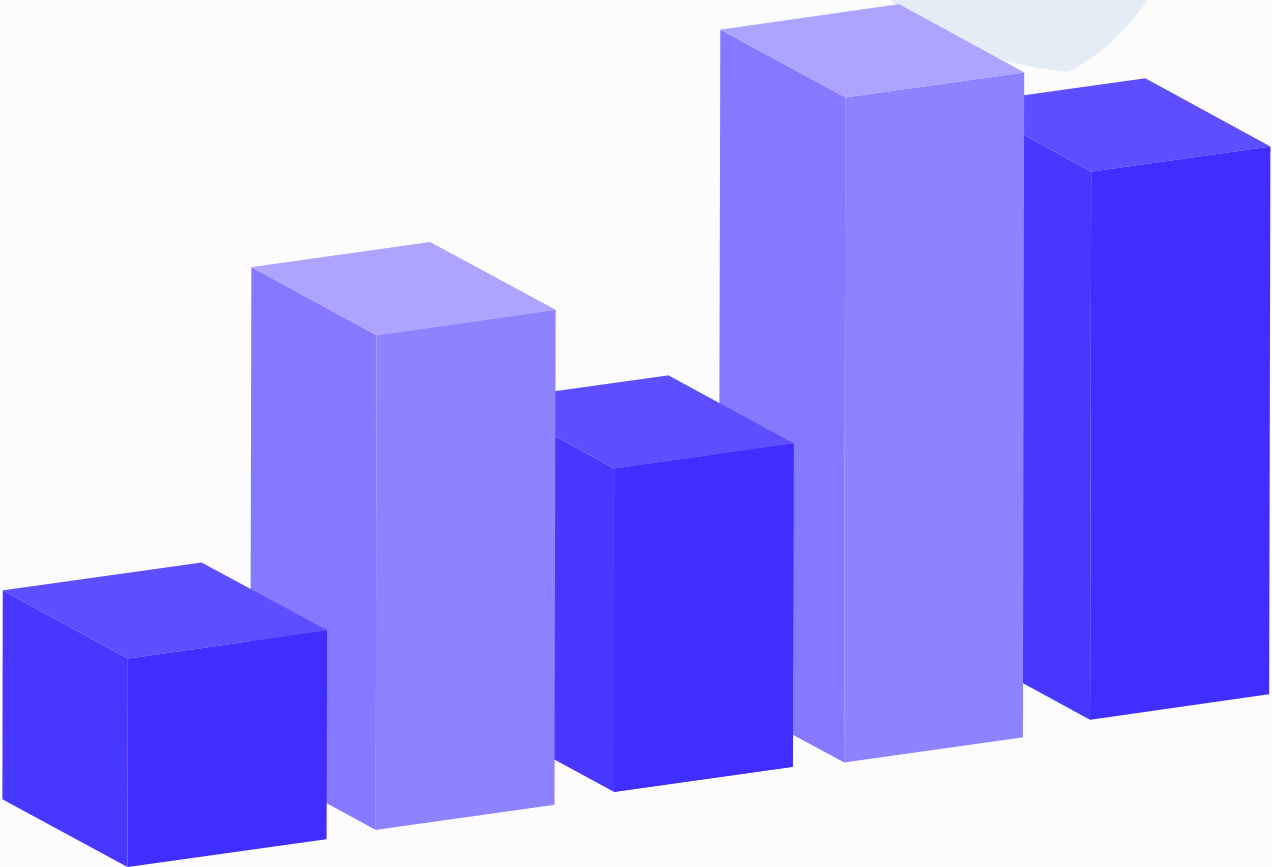
- Data Cleaning & Preprocessing (Python)
- Database Management & SQL Analysis (MySQL)
- Interactive Dashboard Development (Power BI)
- Business Insights & Recommendations



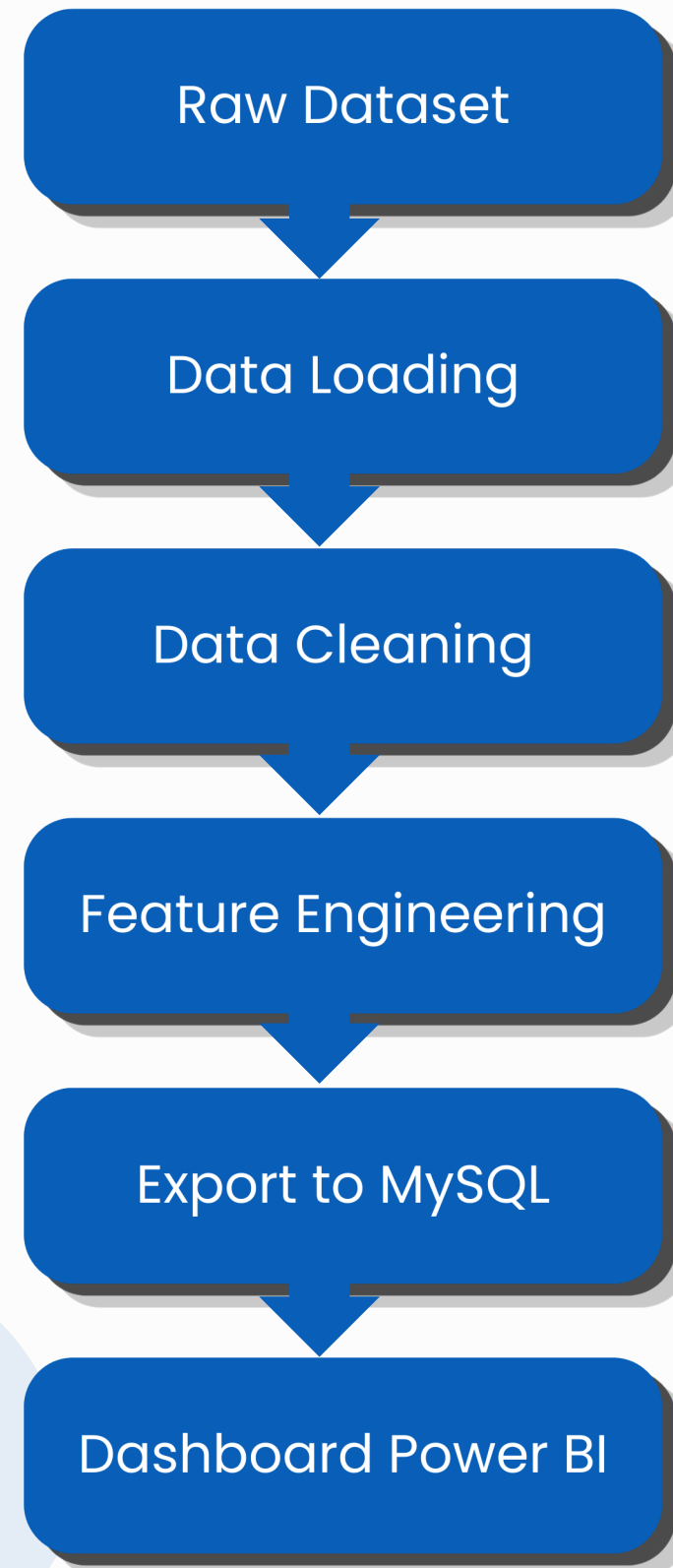
Dataset Overview

Dataset Information

Attribute	Description
Dataset	Customer Shopping Trends
Source	Kaggle
Records	3,900 Customer Transactions
Features	18 Variables
Data Type	Customer Demographics, Purchasing Behavior, Product Information



Data Preparation



Data Loading

- Imported dataset
- Initial exploration

Data Cleaning

- Imputed missing
- Standardized columns
- Removed redundancy

Feature Engineering

- Created Age Group
- Converted purchase frequency to days

Data Export

- Exported clean data to MySQL

SQL Analysis

No.	Business Questions	SQL Technique	Business Purpose
Q1.	Compare revenue by gender	GROUP BY, SUM()	Identify revenue contribution by gender
Q2.	Identify high-value customers using discounts	SUBQUERY, WHERE	Analyze customer spending behavior
Q3.	Find top-rated products	GROUP BY, AVG(), ORDER BY	Evaluate product performance
Q4.	Compare purchase amount by shipping type	GROUP BY, AVG()	Analyze shipping preferences
Q5.	Compare subscribers and non-subscribers	COUNT(), SUM(), AVG()	Evaluate subscription effectiveness
Q6.	Identify products with the highest discount usage	CASE WHEN, GROUP BY	Assess discount strategy
Q7.	Segment customers based on purchase history	CASE WHEN, CTE	Customer segmentation
Q8.	Find top-selling products by category	Window Function, ROW_NUMBER()	Product performance analysis
Q9.	Analyze subscription behavior of repeat buyers	WHERE, GROUP BY	Customer retention analysis
Q10.	Compare revenue across age groups	GROUP BY, SUM()	Demographic analysis

Dashboard



Link: https://github.com/nadyadmy/customer_shopping_behavior_analysis-project/blob/main/customer%20behavior%20dashboard.pbix

Key Findings

1. Young Adults generated the highest revenue.

Young Adult customers contributed the highest total revenue (approximately \$62K), making them the most valuable customer segment.

2. Clothing was the best-performing product category.

The Clothing category generated the highest revenue (approximately \$101K) and recorded the largest number of purchases (1.7K sales), indicating strong customer demand.

3. Subscription adoption remains low.

Only 27% of customers were subscribed, while 73% were non-subscribers. However, SQL analysis showed that subscribers tend to have higher average spending.

4. Customers spend an average of \$59.76 per purchase.

The average purchase amount reached \$59.76, suggesting relatively consistent spending across customer transactions.

5. Customer satisfaction is moderate.

The average review rating was 3.75/5, indicating there is still room to improve customer experience and product satisfaction.

6. Revenue distribution across age groups is relatively balanced.

Although Young Adults generated the highest revenue, the spending differences among age groups were relatively small, indicating a broad customer base.

Business Recommendation

Priority

Recommendation

Expected Business Impact

High	Increase subscription benefits and loyalty incentives to encourage more customers to join the membership program.
High	Prioritize marketing campaigns for the Clothing category, which contributes the highest revenue and sales.
Medium	Develop personalized promotions targeting Young Adult customers.
Medium	Improve product quality and customer service to increase review ratings.
Medium	Maintain sufficient inventory for high-demand products to prevent stock shortages.

Improve customer retention and increase recurring revenue.
Maximize revenue from the highest-performing product category.
Strengthen engagement with the highest-revenue customer segment.
Enhance customer satisfaction and encourage repeat purchases.
Ensure product availability and support continuous sales growth.

Conclusion

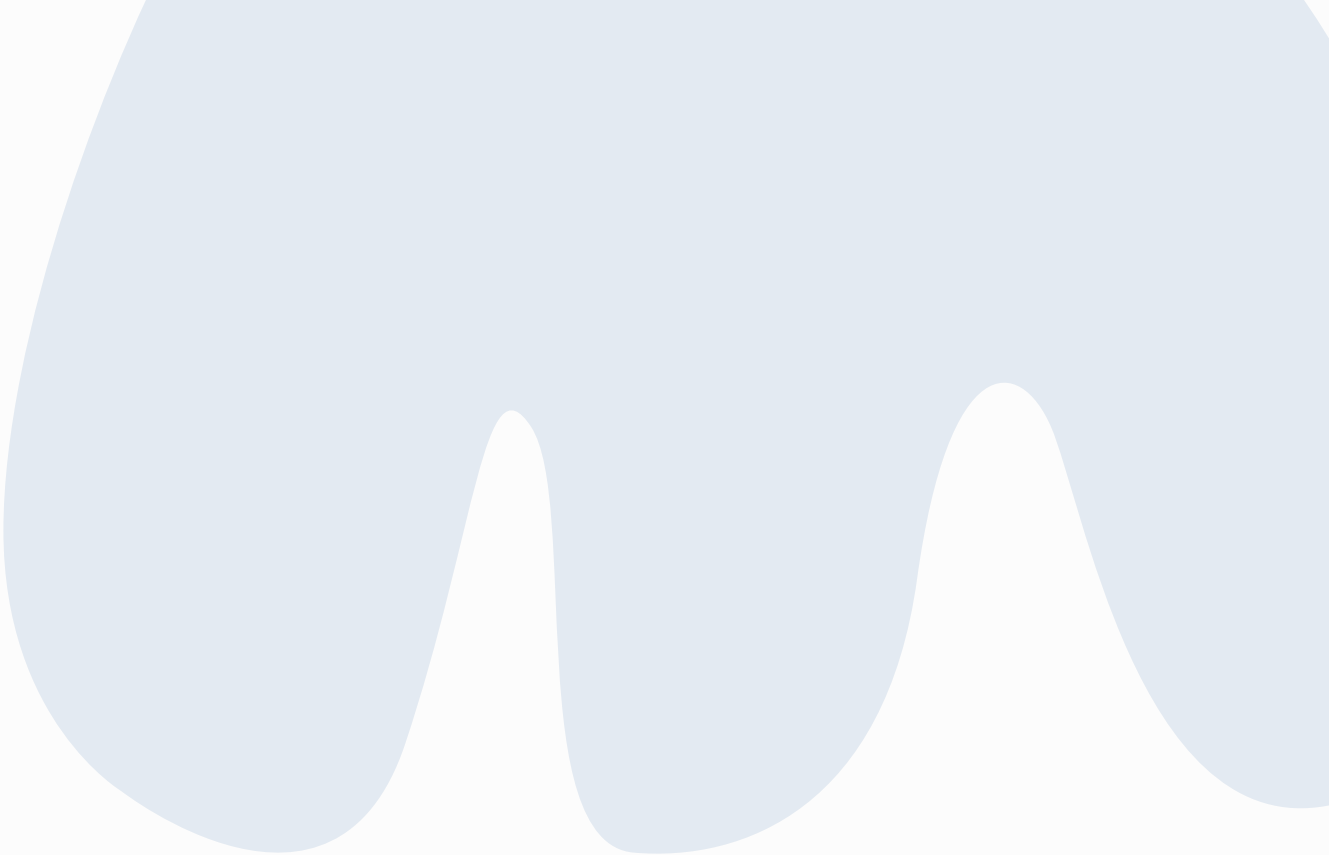
Conclusion

This project successfully analyzed customer shopping behavior using Python, MySQL, and Power BI to transform raw transaction data into meaningful business insights.

The analysis identified key customer segments, purchasing patterns, and product performance, providing actionable recommendations to support marketing strategies, customer retention, and revenue optimization.

Skills Demonstrated

- Python (Data Cleaning & Feature Engineering)
- MySQL (Business Analysis)
- SQL Query Development
- Power BI Dashboard Development
- Data Visualization
- Business Insight & Storytelling



Thank You

